COMPREHENSIVE PROJECT TOPICS LIST

If the purpose is to **get inspired before locking the research problem**, I would not think of computer vision as “YOLO applications.” The field is much broader now: image/video understanding, 3D perception, multimodal reasoning, generative models, anomaly detection, robotics, remote sensing, medical imaging, industrial inspection, and real-time edge systems are all active areas. Recent surveys also show a major shift toward **foundation models, multimodal models, 3D/4D perception, and systems that operate under real-world constraints**, rather than just closed-set image classification. (Queen's University Belfast)

Below is a broad **CV idea map**. You don't need to understand all of it now; use it to discover a direction that makes you think, *“That would actually be interesting to build.”*

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1. The fundamental computer-vision problems

These are the building blocks underneath almost everything.

Image classification

> What is in this image?

- disease classification
- product classification
- species recognition
- material identification
- scene classification

Object detection

> Where are the objects?

- YOLO
- Faster R-CNN
- DETR
- RT-DETR

Applications: people, vehicles, defects, animals, components, weapons, cells, etc.

Semantic segmentation

> Which pixels belong to which class?

Useful for:

- road segmentation
- tumors
- crop regions
- defects
- buildings
- water bodies

Instance segmentation

> Which pixels belong to each individual object?

Example:

```
10 people in a crowd
       ↓
10 individual masks
```

Panoptic segmentation

Combines semantic + instance segmentation.

---

2. Object tracking

Instead of:

```
frame 1 → person
frame 2 → person
frame 3 → person
```

you get:

```
Person #17
   ↓
frame 1
frame 2
frame 3
...
frame 500
```

Areas:

- multi-object tracking
- vehicle tracking
- pedestrian tracking
- animal tracking
- sports tracking
- drone tracking
- surgical-tool tracking

This leads into **trajectory prediction**.

---

3. Human understanding

Much larger than face recognition.

Pose estimation

```
head
shoulder
elbow
wrist
hip
knee
ankle
```

Applications:

- sports analysis
- rehabilitation
- ergonomics
- workplace safety
- exercise coaching
- sign language

Action recognition

Identify:

- walking
- running
- falling
- fighting
- stealing
- cooking
- operating machinery

Action anticipation

Instead of recognizing an action after it happens:

> **Predict what is about to happen.**

That's much more interesting.

---

4. Human behavior understanding

This is where your exam-proctoring idea belongs.

Applications:

- exam cheating
- crowd aggression
- suspicious behavior
- workplace safety
- shoplifting
- falls
- suicide-risk behavior
- abnormal movement
- elderly monitoring

Research directions:

- temporal modeling
- pose-based behavior recognition
- interaction recognition
- group behavior
- anomaly detection
- uncertainty estimation

---

5. Crowd intelligence

Your class already has several topics here.

But the field is much larger than counting people.

Crowd counting

```
How many people?
```

Crowd density estimation

```
Where is the crowd concentrated?
```

Crowd flow

```
Where are people moving?
```

Crowd behavior

```
What is the crowd doing?
```

Crowd anomaly

```
Is something unusual happening?
```

Crowd forecasting

```
Where will the crowd move next?
```

Crowd evacuation

Combine CV + simulation:

```
camera
 ↓
people distribution
 ↓
flow estimation
 ↓
simulation
 ↓
optimal evacuation route
```

That's much more research-oriented.

---

6. Surveillance intelligence

A huge CV domain.

Instead of:

> CCTV records video.

You get:

> **CCTV understands events.**

Examples:

- intrusion detection
- abandoned object detection
- perimeter security
- restricted-zone violations
- unusual movement
- violence detection
- theft detection
- fire/smoke detection
- weapon detection
- loitering detection
- crowd anomalies
- vehicle anomalies

And increasingly:

Video-language models

```
CCTV video
    ↓
Vision-language model
    ↓
"What happened here?"
```

---

7. Video understanding

This is one of the areas I'd strongly recommend exploring.

Instead of processing individual frames:

```
frame → object
```

the system understands:

```
video
 ↓
events
 ↓
relationships
 ↓
actions
 ↓
story
```

Examples:

> "A person entered the restricted area, picked up an object and left."

This leads directly into:

Natural-language video search

> "Find all videos where a red car enters the parking area."

Your class already has this topic.

---

8. Video reasoning + GenAI

A newer direction.

```
Video
 ↓
Vision-language model
 ↓
reasoning
 ↓
answer
```

Questions:

> "Why did the machine stop?"

> "What happened before the accident?"

> "Which worker violated the safety procedure?"

> "Summarize everything unusual that happened between 2–4 PM."

This is where **CV + LLM/VLM** becomes interesting.

Foundation-model research is explicitly moving toward models that combine visual, textual and other modalities and support prompt-driven vision tasks. (Queen's University Belfast)

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9. Industrial computer vision

This is the area you've been considering.

Surface inspection

- scratches
- cracks
- dents
- corrosion
- stains

Assembly inspection

- missing components
- wrong components
- incorrect orientation

- misalignment

PCB inspection

- missing hole
- open circuit
- short
- copper defects

Welding inspection

- cracks
- porosity
- incomplete weld

Textile inspection

- holes
- broken threads
- stains

Semiconductor inspection

- wafer defects
- microscopic defects

Predictive maintenance

```
visual appearance
      ↓
anomaly
      ↓
possible failure
      ↓
maintenance
```

Industrial anomaly detection has developed substantially: recent surveys cover supervised, semi-supervised, unsupervised and reinforcement-learning approaches, while newer work also considers RGB + 3D + multimodal inspection. (ScienceDirect)

---

10. The REALLY interesting industrial problem

Unknown defects

```
Training:

A
B
C

Production:

A ✓
B ✓
C ✓
D ???
```

The model has never seen D.

That's:

> **Open-set / anomaly detection**

This is a very active area and a much better research question than simply "detect defects." (ScienceDirect)

---

11. 3D computer vision

This is a completely different world.

Instead of:

```
RGB image
```

you work with:

```
(x, y, z)
```

3D reconstruction

- buildings
- objects
- factories
- archaeological sites
- human bodies

Point clouds

- LiDAR
- depth cameras
- autonomous vehicles

3D object detection

Used in:

- robotics
- autonomous driving
- AR/VR

3D segmentation

3D anomaly detection

3D scene understanding

Recent robotics research combines SLAM, semantic reconstruction, NeRFs and Gaussian splatting. (ScienceDirect)

---

12. NeRFs and Gaussian Splatting

This is one of the areas I'd investigate purely for inspiration.

You photograph something:

```
      camera 1
         \
camera 2 → object ← camera 3
         /
      camera 4
```

Then reconstruct a 3D scene.

Modern approaches include:

- NeRF
- 3D Gaussian Splatting
- neural rendering

Applications:

- digital twins
- robotics
- VR
- cultural heritage

- real-estate visualization
- autonomous navigation

3D reconstruction from satellite imagery is also beginning to incorporate NeRF and Gaussian splatting. (ScienceDirect)

---

13. Autonomous driving

Probably the biggest CV application area.

Perception

- cars
- pedestrians
- cyclists
- traffic signs
- traffic lights

Lane detection

Drivable-area segmentation

3D object detection

Depth estimation

Tracking

Trajectory prediction

Driver monitoring

Collision prediction

Parking

Night/rain/fog perception

Sensor fusion

```
camera
+
LiDAR
+
radar
 ↓
fusion
 ↓
world model
```

---

14. Vehicle Re-identification

This appears in your class list.

Question:

> Is this the same vehicle seen by Camera A 20 minutes ago?

```
Camera A
   ↓
vehicle embedding
   ↓
database
   ↓
Camera B
   ↓
same vehicle?
```

Research problems:

- occlusion
- viewpoint change
- lighting
- camera shift
- domain generalization

This is significantly more advanced than ordinary object detection.

---

15. Robotics

CV becomes the robot's eyes.

Visual navigation

```
camera
 ↓
understand environment
 ↓
where am I?
 ↓
where should I go?
```

Object manipulation

```
find object
 ↓
estimate pose
 ↓
grasp
```

Visual servoing

Use visual feedback to control the robot.

Robot inspection

A robot moves around a factory and autonomously searches for defects.

Vision-language robotics

> "Find the damaged component and bring it to me."

---

16. Agriculture

Huge field.

Crop disease

Weed detection

Fruit counting

Fruit ripeness

Yield estimation

Plant phenotyping

Weed-vs-crop segmentation

Autonomous harvesting

Soil analysis

Livestock monitoring

Drone-based crop inspection

Recent reviews cover CV across crop monitoring, harvesting, soil/water management and robotic agriculture. (ASCEE Pubs)

---

17. Medical computer vision

One of the biggest fields.

Medical image segmentation

- tumors
- organs
- lesions

X-ray analysis

CT

MRI

Ultrasound

Histopathology

Cell detection

Blood-cell analysis

Retinal imaging

Skin lesions

Surgical vision

Surgical instrument tracking

Medical anomaly detection

---

18. Microscopy

A particularly accessible student research area.

You can work with:

- blood cells
- bacteria
- parasites
- cancer cells
- tissue structures

Problems:

```
cell detection
cell counting
cell tracking
cell classification
cell segmentation
abnormal-cell detection
```

---

19. Remote sensing / satellite CV

Much larger than "traffic congestion from satellite images."

Land-use classification

Building detection

Road extraction

Flood detection

Forest monitoring

Deforestation

Wildfire detection

Glacier monitoring

Crop monitoring

Urban expansion

Disaster assessment

Ship detection

Illegal mining

Change detection

---

20. Satellite change detection

This is particularly interesting:

```
Satellite image 2024
        ↓
       AI
        ↑
Satellite image 2026
        ↓
difference
```

Find:

- new buildings
- destroyed buildings
- deforestation
- floods
- construction
- road changes

This is essentially **temporal computer vision at planetary scale**.

---

21. Multispectral / hyperspectral vision

Instead of RGB:

```
R
G
B
```

you might have:

```
100+ spectral bands
```

Applications:

- agriculture
- mineral detection
- food quality
- pollution
- camouflage
- medical diagnosis
- satellite imaging

This is one of the areas where CV overlaps strongly with remote sensing.

---

22. Low-light / adverse-weather vision

Extremely practical.

Low-light enhancement

Night vision

Rain removal

Fog removal

Snow removal

Motion deblurring

Image denoising

HDR reconstruction

Your class already has:

> Low-SNR satellite image denoising

But the general problem is much broader.

---

23. Image restoration

Given:

```
bad image
 ↓
AI
 ↓
better image
```

Tasks:

- denoising
- deblurring
- super-resolution
- inpainting
- compression artifact removal
- old-photo restoration
- underwater enhancement

---

24. Super-resolution

```
128 × 128
    ↓
AI
```

```
    ↓
512 × 512
```

Applications:

- satellite
- surveillance
- medical imaging
- microscopy
- license plates
- astronomy

A particularly interesting problem is:

> **Can super-resolution actually improve downstream detection?**

rather than simply making an image visually prettier.

---

25. Generative computer vision

Now you enter:

- GANs
- diffusion models
- image generation
- image editing
- synthetic data
- defect generation
- simulation

For your inspection project:

```
normal image
     ↓
synthetic defect generation
     ↓
training data
```

Then:

> Does synthetic data improve real-world inspection?

That's a very usable research question.

---

26. Synthetic data

This is increasingly important.

Instead of collecting:

```
100,000 real images
```

generate:

```
synthetic images
+
simulation
+
augmentation
```

Applications:

- autonomous driving
- robotics

- industrial inspection
- rare-event detection
- medical imaging

---

27. Vision foundation models

This is one of the biggest current shifts.

Instead of training:

```
one model
→
one task
```

you use a pretrained foundation model and adapt it.

Examples of capabilities:

- segmentation
- detection
- image understanding
- visual question answering
- grounding
- tracking
- image editing

Research problems include:

- domain adaptation
- hallucination
- reliability
- evaluation
- explainability
- computational cost
- bias

Foundation-model surveys explicitly identify real-world understanding, evaluation, bias, adversarial vulnerability and interpretability as open problems. (Queen's University Belfast)

---

28. Vision-Language Models

```
IMAGE
 +
TEXT
 ↓
MODEL
 ↓
UNDERSTANDING
```

Examples:

> "What is wrong with this machine?"

> "Which component is missing?"

> "Describe everything unusual in this image."

> "Find the red object near the damaged component."

This can turn CV from:

> **detect**

into:

> **detect + explain + reason**

---

29. Multimodal computer vision

Combine:

```
RGB
+
Depth
+
Thermal
+
Audio
+
Text
+
LiDAR
```

For example:

Factory

```
camera + thermal camera + vibration sensor
```

Surveillance

```
video + audio
```

Autonomous driving

```
camera + LiDAR + radar
```

Medical

```
MRI + clinical text
```

This is a huge research direction.

---

30. Thermal vision

Applications:

- people detection
- fire detection
- industrial overheating
- electrical faults
- night surveillance
- medical monitoring
- wildlife

Interesting problem:

> **RGB fails at night → can RGB + thermal fusion solve it?**

---

31. Event cameras

Very different from normal cameras.

Instead of taking frames:

```
frame
frame
frame
```

event cameras record **changes in brightness asynchronously**.

Useful for:

- high-speed motion
- robotics
- low latency
- low-light
- autonomous systems

This is a more adventurous research direction.

---

32. Vision + audio

For example:

```
video → person falling
audio → impact sound
```

Combine them.

Applications:

- surveillance
- industrial machinery
- accidents
- wildlife
- healthcare

---

33. Vision + sensors

This gets you into **multimodal sensing**.

Example:

```
Camera
+
temperature
+
vibration
+
pressure
```

Then:

> Visual appearance + sensor readings → equipment failure prediction.

This could make an industrial project much more interesting.

---

34. Anomaly detection

Not just industrial.

Image anomaly

> Something looks unusual.

Video anomaly

> Something unusual happened.

Medical anomaly

> This region doesn't look normal.

Cyber-physical anomaly

> Visual state disagrees with sensor state.

Behavioral anomaly

> Person/group behavior is unusual.

Open-set anomaly

> Model sees something it has never learned.

This is a **major umbrella topic**. Recent reviews cover industrial, medical and security applications and distinguish statistical, generative and representation-based approaches. (ScienceDirect)

---

35. Explainable computer vision

Instead of:

```
AI: DEFECT
```

you want:

```
AI: DEFECT

confidence: 94%

reason:
┌──────────────┐
│ highlighted  │
│ region       │
└──────────────┘
```

Methods:

- Grad-CAM
- saliency maps
- attention visualization
- concept-based explanations
- vision-language explanations

This is particularly valuable for:

- medicine
- manufacturing
- safety
- finance

Explainability combined with vision foundation models is itself an active research area. (ScienceDirect)

---

36. Uncertainty-aware vision

Instead of:

```
DEFECT: 98%
```

the model can say:

```
DEFECT: 52%
UNCERTAIN
```

This is important in safety-critical applications.

Research:

- uncertainty estimation
- calibration
- confidence estimation
- selective prediction
- abstention

Your class's:

> **Uncertainty-Aware Vehicle Re-ID**

is an example.

---

37. Adversarial computer vision

Another very different direction.

Adversarial examples

Tiny perturbation:

```
normal image
    ↓
imperceptible modification
    ↓
AI gets fooled
```

Physical adversarial attacks

Put something physical into the scene:

```
adversarial patch
     ↓
camera
     ↓
YOLO
     ↓
wrong detection
```

Defense

Then:

> How do we make vision models robust against adversarial attacks?

That's why your class has the adversarial-patch topic.

---

38. Privacy-preserving computer vision

Instead of storing raw CCTV:

```
camera
 ↓
edge processing
 ↓
extract event
 ↓
discard video
```

This creates interesting questions:

- Can we detect events without storing identities?
- Can we blur faces while retaining behavior information?
- Can anomaly detection work without persistent video?
- Can processing happen entirely on-device?

That's a much deeper topic than ordinary surveillance.

---

39. Edge AI

Take the model out of the cloud.

```
Camera
 ↓
Jetson / Raspberry Pi / NPU
 ↓
AI
 ↓
decision
```

Research constraints:

- latency
- memory
- power
- model compression
- quantization
- pruning
- distillation

This gives strong engineering/resume value.

---

40. TinyML / extremely lightweight vision

Even more constrained:

```
microcontroller
+
camera
+
tiny neural network
```

Question:

> Can useful vision run on hardware costing only a few dollars?

Very different from GPU-based deep learning.

---

41. Model compression

Take:

```
2 GB model
```

and make:

```
100 MB
```

while retaining accuracy.

Techniques:

- pruning
- quantization
- knowledge distillation
- low-rank adaptation

This is a great **AI + systems** research direction.

---

42. Federated computer vision

Imagine ten cameras/factories/hospitals.

Instead of sending their images to a central server:

```
Camera A ─┐
Camera B ─┼── local training
Camera C ─┘
       ↓
model updates
       ↓
central model
```

No raw images need to leave the site.

Applications:

- medical imaging
- surveillance
- manufacturing
- mobile vision

---

43. Continual learning

A factory changes.

New product:

```
Model trained in January
        ↓
February → new component
        ↓
March → new defect
        ↓
April → new machine
```

Can the model continuously learn without forgetting old knowledge?

That's:

> **Continual / lifelong learning**

Very interesting for automated inspection.

---

44. Domain adaptation / domain generalization

Train:

```
Factory A
```

Deploy:

```
Factory B
```

without retraining from scratch.

Or:

```
day → night
sunny → rainy
camera A → camera B
lab → real world
simulation → reality
```

This is one of the most practical research problems in CV.

---

45. Cross-domain robustness

Ask:

> How much does a model degrade when the environment changes?

This is an excellent project because the experiment is straightforward:

```
Train
 ↓
Dataset A
 ↓
test A
test B
test C
test D
```

Then investigate techniques that recover the lost performance.

---

46. Digital twins + computer vision

Your class has this too.

```
Real factory/city
       ↕
Digital twin
       ↕
AI
```

Applications:

- traffic
- manufacturing
- buildings
- logistics
- cities
- infrastructure

CV provides the perception layer.

---

47. Infrastructure inspection

Very underrated.

Roads

- potholes
- cracks
- lane deterioration

Bridges

- cracks

- corrosion
- deformation

Railways

- rail cracks
- track geometry
- vegetation
- obstacles

Buildings

- structural cracks
- facade damage

Power lines

- damaged components
- vegetation encroachment

Pipelines

- corrosion
- leakage indicators

Your class's **RailSafe** is an example.

---

48. Drone computer vision

Put a camera on a UAV.

Applications:

- infrastructure inspection
- agriculture
- disaster response
- search and rescue
- wildlife
- construction monitoring
- mining

Then add autonomy:

```
drone
 ↓
detect anomaly
 ↓
navigate toward it
 ↓
inspect
 ↓
report
```

That's a much richer system.

---

49. Construction-site vision

Applications:

- helmet/PPE detection
- worker tracking
- unsafe behavior
- equipment monitoring
- progress estimation

- material counting
- site mapping
- hazard detection

---

50. Retail computer vision

Applications:

- shelf monitoring
- inventory detection
- checkout automation
- customer movement
- queue estimation
- product recognition
- misplaced products
- planogram compliance

---

51. Sports computer vision

You already have cricket technique in your list.

But:

Player tracking

Ball tracking

Pose estimation

Shot classification

Tactical analysis

Performance prediction

Injury-risk estimation

Automated highlights

Referee assistance

3D motion reconstruction

You could even do:

> **Estimate cricket shot biomechanics using monocular video.**

---

52. Education

Beyond proctoring:

- handwriting recognition
- board digitization
- gesture recognition
- classroom attention
- chalk/marker tracking
- mathematical equation recognition
- automated diagram understanding
- sign-language learning
- handwriting disorder detection

Your class has **GraphoLens** and chalk-stroke extraction.

53. Document vision

Huge field.

- OCR
- handwriting recognition
- table extraction
- mathematical-expression recognition
- document layout analysis
- signature detection
- invoice understanding
- form understanding

Then combine it with LLMs:

```
document image
 ↓
vision model
 ↓
structured information
 ↓
LLM reasoning
```

---

54. Accessibility

Computer vision can become assistive technology.

Examples:

- scene description for blind users
- obstacle detection
- sign-language translation
- lip reading
- object identification
- indoor navigation
- text reading
- facial-expression assistance

---

55. Environmental computer vision

Huge possibilities:

- plastic detection
- marine debris
- wildfire
- pollution
- illegal dumping
- deforestation
- wildlife monitoring
- invasive species
- glacier change
- coral reef health

---

56. Wildlife vision

```
camera trap
 ↓
AI
```

```
 ↓
species
 ↓
individual
 ↓
behavior
```

Problems:

- rare species detection
- animal re-identification
- population estimation
- migration tracking
- disease detection

---

57. Underwater vision

Very different constraints:

- low visibility
- color distortion
- scattering
- motion
- limited labeled data

Applications:

- coral monitoring
- marine debris
- underwater archaeology
- fish counting
- infrastructure inspection

---

58. Astronomy / space vision

- galaxy classification
- asteroid detection
- crater detection
- exoplanet detection
- solar activity
- satellite damage
- space debris detection

---

59. Human-computer interaction

Vision as an interface:

```
camera
 ↓
hand
 ↓
gesture
 ↓
computer command
```

Applications:

- gesture control
- AR/VR
- sign language

- touchless interfaces
- eye tracking

---

60. Augmented / virtual reality

Computer vision handles:

- SLAM
- object tracking
- hand tracking
- body tracking
- depth estimation
- scene reconstruction
- spatial understanding

---

61. Generative + perception systems

This is an especially interesting modern direction:

```
Vision model
     ↓
understand scene
     ↓
Generative model
     ↓
simulate possible future
```

Applications:

- autonomous driving
- robotics
- forecasting
- video prediction

---

62. Future prediction

Rather than:

> "What is happening?"

ask:

> **"What happens next?"**

Examples:

```
pedestrian movement
vehicle trajectory
crowd flow
machine failure
human action
weather evolution
```

Your class already has:

> Multi-Object Tracking and Future Trajectory Prediction

because this is a genuine research direction.

---

63. Video prediction

```
Frames 1–20
    ↓
AI
    ↓
predict frames 21–40
```

Potential uses:

- autonomous driving
- surveillance
- robotics
- weather
- human motion

---

64. World models

This is the extreme end of the spectrum.

Instead of merely detecting objects:

```
car
person
road
```

the model learns something closer to:

```
"What is happening in this environment,
what caused it, and what might happen next?"
```

This connects computer vision with:

- multimodal foundation models
- robotics
- simulation
- planning
- autonomous agents

---

65. A useful way to think about your project

Don't start with:

> **"What application should we build?"**

Start with:

What CV capability do we want to investigate?

Then choose the application.

For example:

CV research capability	Possible application
Object detection	manufacturing
Segmentation	medical
Tracking	traffic
Re-ID	surveillance
Anomaly detection	factory
Open-set recognition	inspection
Domain adaptation	satellite
Few-shot learning	rare defects

CV research capability	Possible application
Temporal reasoning	exam hall
Trajectory prediction	crowd
3D reconstruction	robotics
Multimodal fusion	autonomous driving
Vision-language reasoning	CCTV
Synthetic data	manufacturing
Adversarial robustness	surveillance
Explainability	medical
Uncertainty	safety
Edge AI	drones
Continual learning	factories
Federated learning	hospitals
NeRF/3DGS	digital twins
Hyperspectral CV	agriculture
Thermal CV	industrial safety
Video anomaly	security
Future prediction	traffic

This is the key insight for your situation.

Your assigned title:

> **AI-Based Computer Vision for Automated Inspection Using Deep Learning**

doesn't force you to build a conventional defect detector.

You can interpret **“inspection” as an intelligent perception/decision task**.

For example:

> Inspect a factory component for unknown defects.

or:

> Inspect a railway track for emerging structural anomalies.

or:

> Inspect an assembly for missing/misplaced components.

or:

> Inspect a crowd for dangerous behavioral transitions.

or:

> Inspect satellite imagery for changes.

or:

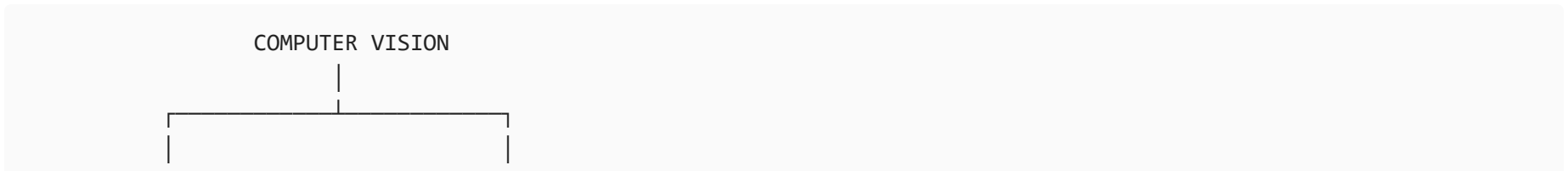
> Inspect CCTV video for anomalous events.

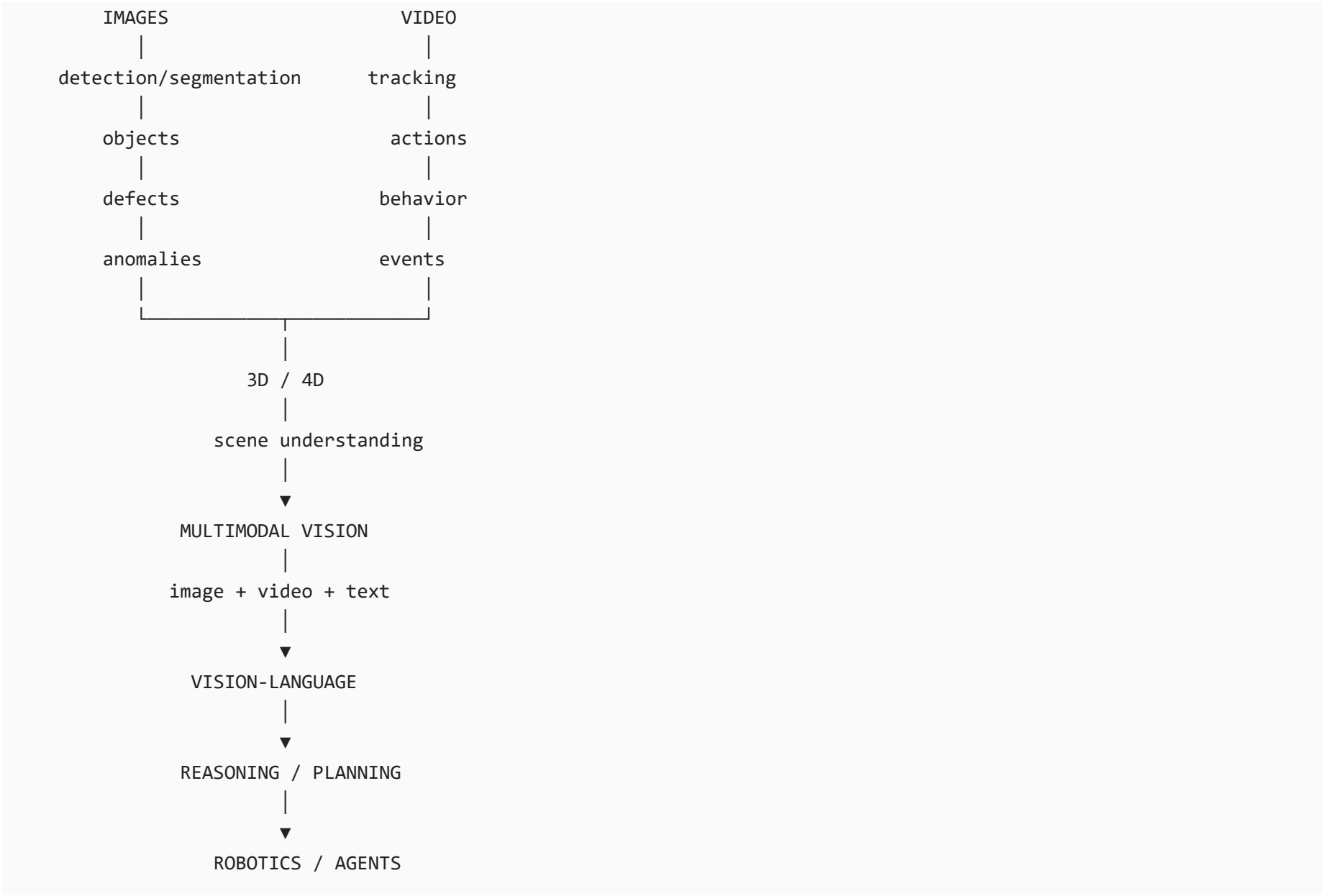
or:

> Inspect machinery using RGB + thermal imagery.

The technology progression I'd keep in mind

There's a useful evolutionary ladder:





And underneath all of this are the **research constraints**:

```
Limited data
Class imbalance
Unknown classes
Domain shift
Occlusion
Low light
Small objects
Real-time inference
Edge hardware
Privacy
Uncertainty
Explainability
Adversarial robustness
Continual learning
Multimodal fusion
```

That bottom layer is where I would look for your unique project.

For example, don't merely choose:

> **"Industrial defect detection."**

Choose:

> **"Industrial defect detection under unseen defect classes and limited defective samples."**

Don't merely choose:

> **"Crowd monitoring."**

Choose:

> **"Early detection of dangerous crowd behavior from temporal interactions rather than static crowd density."**

Don't merely choose:

> **"Satellite image analysis."**

Choose:

> **"Robust satellite change detection under seasonal/domain variation."**

Don't merely choose:

> **"Exam proctoring."**

Choose:

> **"Multi-person temporal interaction anomaly detection under occlusion."**

That transformation—from **application** → **research constraint** → **measurable problem**—is what will make your literature survey useful rather than just a collection of 10 papers.

For your next step, I would build a **large candidate matrix of ~30–50 project ideas**, each with **(1) research gap, (2) 10+ paper potential, (3) public datasets, (4) synthetic-data possibility, (5) difficulty, and (6) resume tech stack**, and then narrow it to 3–5 before we start the actual literature survey.